Title of the project :- Replace and modernize legacy VPN architectures with a cloud-native Zscaler Zero Trust solution.

Overview :-

Traditional Virtual Private Network (VPN) architectures have been widely used by organizations to allow remote users to access internal resources. However, legacy VPN solutions often provide broad network access once a user is authenticated. This means hat users connected through a VPN may gain access to large portions of the internal network, increasing the risk of cyber attacks such as lateral movement and data breaches.

Modern cybersecurity strategies are movingtoward the Zero Trust security model, which follows the principle of "never trust, always verify." Instead of granting full network access, Zero Trust architecture allows users to access only specific applications based on identity, device posture, and scurity policies.

One of the leading solutions that implements Zero Trust Network Access (ZTNA) is developed by . Zscaler provides a cloud-native security platform that securely connects users directly to applications without exposing the entire network infrastructure.

Unlike traditional VPN systems that create a network tunnel between users and the internal network, Zscaler establishes secure connections between users and specific applications through a cloud broker. This architecture hides internal applicatins from the internet and ensures that only authorized users can access them.

By replacing legacy VPN solutions with Zero Trust architectures, organizations can significantly reduce their attack surface and prevent attackers from moving laterally within the network after gaining initial access. This approach also improves security for remote work environments and cloud-based applications.

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List of Vulnerability Table

S.No	Vulnerability Name	CWE/CVE No
1	Weak Authentication in VPN Systems	CWE-287
2	Network Exposure Through VPN	CWE-284
3	Lateral Movement After Compromise	CWE-668
4	Security Misconfiguration	CWE-16

Vulnerability Report:-

Legacy VPN infrastructures often expose internal networks to remote users once authentication is completed. If an attacker compromises a user account or dvice, they may gain access to multiple systems within the network. This situation increases the risk of lateral movement where attackers move between systems to escalate privileges and access sensitive data.

Security misconfigurations and weak authentication mechanisms further increase the risk of unauthorized access. Traditional VPN architectures also expose network ports and internal resources that may be discovered and exploited by attackers.

Understandingthese vulnerabilities is importantin designingsecure access solutions that limit network exposure and protect internal applications.

Business Impact

Securityvulnerabilities in traditional VPN infrastructures can lead to significant business risks including data breaches, financial loss, operational disruption, and reputational damage. Attackers who gain access to internal networks through compromised VPN credentials may move laterally across systems and access sensitive information.

Implementing Zero Trust Network Access solutions such as elps

organizations reduce these risks by restricting user access to specific applications instead of the entire network. This significanly minimizes the attack surface and improves overall cybersecurity resilience.